

# Alaa Malabeh

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## EDUCATION

### University of California, Irvine

Software Engineering, B.S.

Dean's Honor List Recipient

Relevant Coursework: Python, C/C++, Java, Data Structures/Algorithms Implementation & Analysis, UI/UX Software Design, Requirements Engineering, Software Testing & Quality Assurance, Database Management, Boolean Logic, Engineering AI Software, Ubiquitous Computing, AWS

Cumulative GPA: 3.5

Sept. 2022 - Jun. 2026

## PROFESSIONAL EXPERIENCE

### Panasonic Avionics Corporation - Software Engineering/DevOps Intern

Jun. 2025 - Sept. 2025

- Built and maintained GitLab CI/CD pipelines, improving API deployment reliability and reducing manual deployment overhead across environments.
- Developed a Python automation tool using pandas and openpyxl that automated file creation and categorization, saving ~40 hours/month, utilizing Selenium and Playwright for automation.
- Integrated automated testing and deployment workflows for GraphQL APIs, improving release consistency and defect detection.
- Designed reusable GitLab CI templates and optimized job dependencies, enabling scalable CI/CD across multiple services.

### Learning Assistant (LA) - Intro to Software Engineering

Aug 2024 - Dec. 2024

- Guided students in creating requirements documents and UML diagrams, providing foundational support in software engineering concepts.
- Assisted students during lectures and discussion sections, answering questions and clarifying concepts in real time.
- Collaborated with the professor and Teaching Assistants (TAs) in weekly meetings to align teaching strategies and improve student outcomes.
- Earned the Certified Learning Assistants Program (CLAP) certification, demonstrating advanced skills in academic support and student engagement.

### Grader/Reader - Intro to Software Engineering

Sept. 2025 - Dec. 2025

- Evaluated student assignments, projects, and exams for accuracy, code quality, and adherence to software engineering principles.
- Proctored quizzes and maintained academic integrity during assessments.
- Collaborated with the course instructor and attended weekly meetings to discuss grading rubrics, course progress, and student learning outcomes.

### UCI Science Library - Student Check Out Desk & Copy Services Assistant

Apr. 2024 - Jun. 2026

- Delivered high-quality support by responding to patron inquiries over the phone and in person, ensuring satisfaction with library services.
- Facilitated access to digital resources, including online textbooks, reserves, and internet services, enhancing the user experience.

## RESEARCH EXPERIENCE

### AI/LLM Education Impact Research - Research Assistant

Mar. 2024 - Present

- Analyzed 1,200+ UCI student survey responses using one-hot encoding and Cramer's V to identify data relationships between AI usage and long-term impact on educational outcomes.
- Built tooling to automate survey categorization and preprocessing.
- Contributed to research findings used in academic presentations and draft publications using Excel and Pandas.
- Engaged in team discussions to refine methodologies and overcome research challenges.
- Co-authored a peer-reviewed paper accepted for publication at the International Conference on Computer Science and Education (ICCSE), 2026.

## PROJECT EXPERIENCE

### Gamified Psychomotor Vigilance Test (PVT) App - Capstone Project, Project Manager and Developer

Jan. 2026- Present

- Awarded 2nd Place at the National Capstone Design Conference Competition and 3rd Place at the UCI ICS Project Expo.
- Designing and implementing a Unity and C# based mobile application to deliver research-grade cognitive testing for pediatric and adult users.
- Built and iterated on 30+ gamified task variants while preserving standardized PVT mechanics.
- Developed core game mechanics, including reaction timing, input handling, and validation logic for Psychomotor Vigilance Tasks in 3D.
- Designed low-fidelity and high-fidelity game mockups to guide development and user experience.
- Converted stakeholder research goals into defined system constraints, risks, and future extensibility to support long-term research use.

### Roommate Court- Full-Stack Development, <https://roommate-court.web.app/>

Mar. 2026 - Jun. 2026

- Designed and developed a web-based dispute resolution platform that transforms roommate conflicts into a structured "courtroom" workflow.
- Built dynamic React components to support case creation, evidence submission (as photo/text uploads), and real-time UI updates for multi-user interactions.
- Implemented a jury based voting system and integrated logic for an automated AI judge to generate verdicts based on user input.
- Developed features for case tracking and historical records, enabling users to view past disputes, rulings, and accountability within a shared household using Firebase.
- Created low and high-fidelity mockups and translated them into functional UI flows, improving usability and ensuring consistency across features.

### Menstrual Product Tracker App- Full-Stack Development, Hackathon

May. 2025

- Designed and developed a menstrual product locator app using Ionic and Angular to help users find free hygiene products across campus and community locations.
- Integrated the Google Maps API to display real-time location markers and interactive navigation within the app.
- Collaborated in a fast-paced hackathon environment, applying rapid prototyping, UI iteration, and user-centered design principles to deliver a functional demo within time constraints.

### Spotify Music Browser- Full-Stack Development

Feb. 2025

- Developed a full-stack music browsing application using Angular and Node.js.
- Integrated Spotify's API via OAuth authentication, making HTTP requests to fetch and display artists, albums, and tracks dynamically.
- Implemented UI components including search functionality, carousels for browsing, and optimized user experience by applying responsive design and structuring data efficiently in TypeScript.

## SKILLS

**Technical skills:** Python, Java, C++, SQL, HTML, CSS, JavaScript, Figma, Agile Development, Ionic, Angular, YAML, Unity, React, Bash

**Soft Skills:** Event Coordination Experience, Conflict Resolution, Strong Communication, Leadership Experience

**Tools:** Microsoft Office (Excel, PowerPoint), Pandas